

Databases and Web Programming Week 2 Lab

This lab sheet is designed to help you understand and apply the principles of database normalization, specifically focusing on the first three normal forms (1NF, 2NF, and 3NF).

Learning Objectives

- Understand the concept of data redundancy and update anomalies.
- Identify and correct violations of 1NF, 2NF, and 3NF.
- Decompose a denormalized table into a set of normalized tables.
- Create a logical database schema based on the normalized tables.

Exercise 1: First Normal Form (1NF)

Examine the Students table below. Identify any violations of 1NF and propose a new, normalized structure.

Student ID	Name	Course	Instructor	Course Grade
100	Alice	Maths, Physics	Smith, Jones; Doe, John	A, B
102	Bob	Physics	Doe, John	A
104	Alice	English, Art	Doe Jane	A, A
108	Bob	Art, Maths	James, Joe; Doe, Jay	B, B

Questions:

1. Which columns violate 1NF? Why?
2. How would you restructure this table to be in 1NF? Show your proposed table structure and some sample data.

Exercise 2: Second Normal Form (2NF)

Consider the OrderDetails table below, which has a composite primary key of (OrderID, ProductID).

OrderID (PK)	ProductID (PK)	ProductName	OrderDate	CustomerName	Quantity	Price
1001	501	Laptop	10-09-2025	John Doe	1	1200
1001	502	Mouse	10-09-2025	John Doe	1	25
1002	501	Laptop	16-01-2025	Jane Doe	2	3200

Questions:

1. What is the composite primary key of this table?
2. Identify which columns are functionally dependent on only part of the primary key.
3. Explain the data redundancy and update anomalies present in this table.
4. Decompose this table into new tables that satisfy 2NF. Clearly state the primary and foreign keys for each new table.

Exercise 3: Third Normal Form (3NF)

Examine the `Employees` table below, which is currently in 2NF.

EmployeeID (PK)	EmployeeName	DepartmentID	DepartmentName	DepartmentLocation
E001	Alice	D1	Sales	London
E002	Bob	D2	IT	New York
E003	Charlie	D1	Sales	London

Questions:

1. What is the primary key?
2. Identify the transitive dependency in this table.
3. Explain the potential issues (anomalies) that could arise from this dependency.

4. Decompose this table into tables that satisfy 3NF. Define the primary and foreign keys for each new table.